Information Retrieval and Web Analysis

Final Project PART 1

Indexing and Evaluation

Serena El-Hajj & Souad Ibrahim El Husseini & Guillermo

November 3, 2026

Tag: IRWA-2026-part-1

GitHub: https://github.com/Souad-El-Husseini/irwa-search-engine

AI Use Declaration

- ***Data Preparation:*** *ChatGPT helped us understand the implications and trade-offs of different approaches to handling the fields in Step 3.*
- ***EDA:*** *We used ChatGPT to help develop and improve some of the Python pseudocode for statistical analysis and visualizations.*
- ***Debugging:*** *ChatGPT assisted us in identifying errors, debugging our code, and verifying intermediate results.*

PART 1: Data Preparation:

1.1-Dataset Inspection and Text Preprocessing

Firstly, our main goal was to understand the structure of our dataset and identify any issues that could affect preprocessing. That is why, before implementing our search engine, we started examining the fashion-product dataset.

Dataset shape: (28080, 14) (From the table)

The dataset contains 28,080 products and 14 original attributes, which include: product titles, descriptions, brands, categories, prices, ratings, and stock availability. We checked for missing values, empty strings, duplicate rows, and duplicate product IDs.

Results came out, and no duplicates were found. Additionally, all 28,080 products had unique identifiers.

However, we noticed that some attributes contained missing values. The most significant case was the description field, with 11,153 missing entries. Missing values were also present in brand (2,009) and seller (1,643).

To handle this, we created a separate working copy of the dataset and replaced missing values in textual fields with empty strings. This allowed us to simultaneously apply our preprocessing functions without encountering errors and preserve the original dataset and its information.

Text Preprocessing Pipeline

We implemented our preprocessing pipeline using Python and the Natural Language Toolkit (NLTK). The main steps were:

1. Lowercasing: We converted all text to lowercase so that words such as "Women" and "women" would be treated as the same term.
2. Tokenization and punctuation removal: We used NLTK's RegexpTokenizer with the regular expression [a-z0-9]+ to split text into individual tokens. This also removed punctuation while preserving letters and numbers.

3. Stop-word removal: We removed common English stop words using NLTK's stop-word list. Words such as "a", "of", and "for" generally contribute little to identifying relevant fashion products.
4. Stemming: We applied the Porter Stemmer to reduce related word forms to a common stem. (eg: "pants" becomes "pant").This allows different forms of a word to be matched more consistently.

An example of our process:

Original: "A pair of pants for women"

After preprocessing: ['pair', 'pant', 'women']

Throughout our work, we tested each operation individually to see its effects and then, in order to simplify preprocessing for future steps, we combined them into a single preprocess_text() function. We then applied this function to the title and description fields as required in the instructions.

1.2 Preserving Product Information

Preprocessing is necessary for our search engine to search efficiently; however, when the search is done, our search engine should be able to display meaningful information about retrieved products,

For this reason, we decided to keep the original product attributes unchanged and create additional processed representations for indexing.

Therefore, we verified that all 14 required attributes were preserved:

pid, title, description, brand, category, sub_category, product_details, seller, out_of_stock, selling_price, discount, actual_price, average_rating, and url.

This approach allows us to use processed tokens for searching while retaining the original product titles, descriptions, prices, and URLs for displaying search results.

1.3 Handling Additional Textual Fields

Our dataset does not only contain titles and descriptions. It has five other potentially useful textual attributes: brand, category, sub_category, seller, and product_details.

We decided to preserve the textual fields separately instead of merging them into one.

This way, our inverted index will not only identify which products contain a term from the customer's search query but also determine in which field that term appears.

For example, the word "cotton" could appear in a product description or in its product details. Keeping these fields separate allows us to distinguish between those occurrences.

This could also support future field-specific ranking improvements, meaning that we could give different importance to matches depending on the field in which they occur. For example, a product containing a query term in its title could receive a higher weight than a product containing the same term only in its description.

We apply the same preprocessing function to brand, category, sub_category, and seller so we can stay consistent with the query preprocessing.

Advantages and limitations:

Merging all fields would provide a simpler indexing structure and would allow terms to be searched without considering their original locations. However, it would also remove useful field information and make it more difficult to distinguish highly relevant matches from general ones.

Our separate-field approach preserves more information and supports future field-specific retrieval, but it also introduces additional complexity when building and processing the inverted index.

The dataset contains several distinct types of textual information; therefore, this trade-off is worthwhile.

Processing the product_details Attribute:

Unlike the other textual attributes, product_details contain structured information stored as a string representing a list of dictionaries.

For example, a product may contain attributes such as:

```
[
   {"Style Code": "1005COMBO2"},
   {"Closure": "Elastic"},
   {"Fabric": "Cotton Blend"},
   {"Color": "Multicolor"}
]
```

We used Python's ast.literal_eval() function to convert these strings into Python data structures. We then extracted both the attribute names and their values before applying our text preprocessing function.

This allowed us to preserve meaningful terms such as "fabric", "cotton", "color", and "multicolor", while excluding the brackets, quotation marks, and other formatting characters.

1.4 Handling Numerical and Boolean Attributes

The dataset also contains numerical and Boolean attributes that require a different preprocessing strategy:

selling_price, actual_price, discount, average_rating, and out_of_stock.

We decided not to include these attributes as textual terms in the inverted index and decided to convert prices, discounts, and ratings into numericalvalues to support future filtering and analysis.

That is because users may want to filter products by price, discount, rating or availability, and in this case numerical comparisons are more useful than text terms. (eg: A customer wants to find products that cost less than 200 euros) The original values are preserved, while missing or invalid values remain missing rather than being replaced with zero.

We therefore created additional columns:

Original attribute	Processed attribute
selling_price	selling_price_num
actual_price	actual_price_num
discount	discount_pct
average_rating	average_rating_num
out_of_stock	out_of_stock_bool

We removed formatting characters from prices, such as thousands separators, and converted the values to numerical types.

Then,for discounts, we extracted the numerical percentage from strings such as "69% off".

We converted product ratings into numerical values where necessary, while preserving stock availability as a Boolean attribute.

1.5 Exporting the Preprocessed Data

After preprocessing the textual fields, we organize the resulting tokens into a separate dataset for indexing.

Each product keeps its unique product ID, and the seven textual fields are stored separately. The original product information is preserved in fashion_products.csv for displaying search results.

By separating the indexing representation from the original product information, we established a consistent connection between preprocessing, retrieval, and the later evaluation stages.

Overall, our data preparation process aimed to improve text consistency, and preserve useful product information.

PART 2: Exploratory Data Analysis:

Exploratory Data Analysis (EDA) allows us to better understand the characteristics of our fashion product dataset before using it for information retrieval.

The dataset we are working on contains 28,080 products, and we will be exploring both their textual information and their numerical and categorical attributes. To identify the patterns that may influence the indexing and retrieval of products, we will use descriptive analysis and visualizations.

2.1 Word Count distribution:

We first started analyzing the length of the products' titles and descriptions. Therefore, we counted the number of tokens that remained after preprocessing.

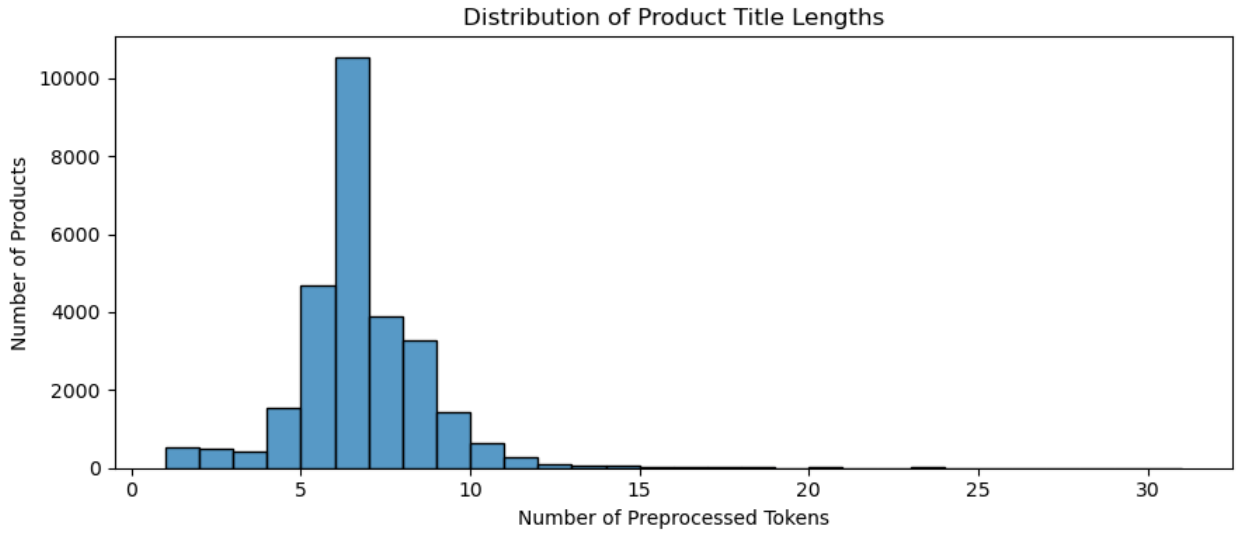
As we can see from the table provided below:

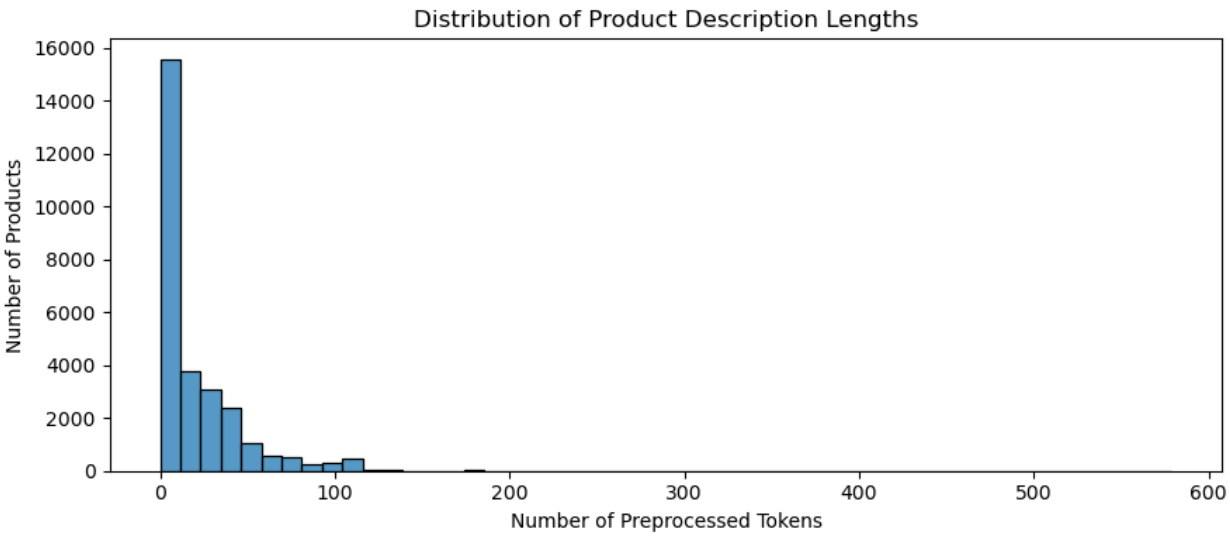
Product titles are relatively short, with an average of 6.27 tokens and a median of 6. Also, we can see that they have a low standard deviation of 1.98, which indicates similarity in their lengths.

Descriptions are generally longer with an average of 18.84 token with a median of only 9. We can also notice that the descriptions are much more variable than the titles, with a standard deviation of 26.72 and a maximum of 579 tokens for one of the products.

We also observed that approximately 39.7% (39.72 %) of the dataset has missing descriptions, which will be represented by empty lists after preprocessing.

This observation reinforces our decision to index multiple fields separately. That way, if a product has a missing description, it does not mean it is irrelevant during indexing, and it could also be potentially “picked” based on its other fields.




Distribution of Product Description Lengths
Number of Products
16000
14000
12000
10000
8000
6000
4000
2000
0
0
100
200
300
400
500
600
Number of Preprocessed Tokens

2.2-Vocabulary Size and Word Frequency Analysis:

Next, we analyzed the preprocessed vocabulary of our dataset by calculating the number of unique terms across the seven textual fields used for indexing.

As we can see from our results, the size of the total vocabulary is 21073 unique terms, with 738 of them in the titles and 4697 in the descriptions. It should be kept in mind that the total vocabulary size includes all 7 text fields; that is why summing the unique terms of the titles and descriptions would be less than 21073.

We also examine the most frequent terms (e.g., shirt: 15838 occurrences, women:13146, etc.) This is related to our theory classes, as frequently occurring terms have less impact than more specific ones.

Most frequent terms in Titles

Term	Frequency	Term	Frequency	Term	Frequency
shirt	15,838	round	8,301	black	3,503
women	13,146	print	8,123	casual	2,927
men	13,060	pack	4,839	slim	2,893
neck	11,952	blue	4,378	polo	2,668
solid	9,246	fit	3,923	collar	2,427

Most frequent terms in Descriptions

Term	Frequency	Term	Frequency	Term	Frequency
shirt	18,321	casual	6,034	design	4,301
cotton	10,139	fit	5,831	sleev	4,221
wear	9,353	made	5,643	qualiti	4,087
comfort	8,836	fabric	5,516	men	3,775
look	6,413	print	4,888	women	4,424

2.3- Average Sentence Length

Next, we analyzed the average sentence length of product descriptions to better understand how the products are described in our dataset.For this analysis, we used the original descriptions before preprocessing; punctuation is needed to identify sentence boundaries.

As we can see from our table below, we can observe that:

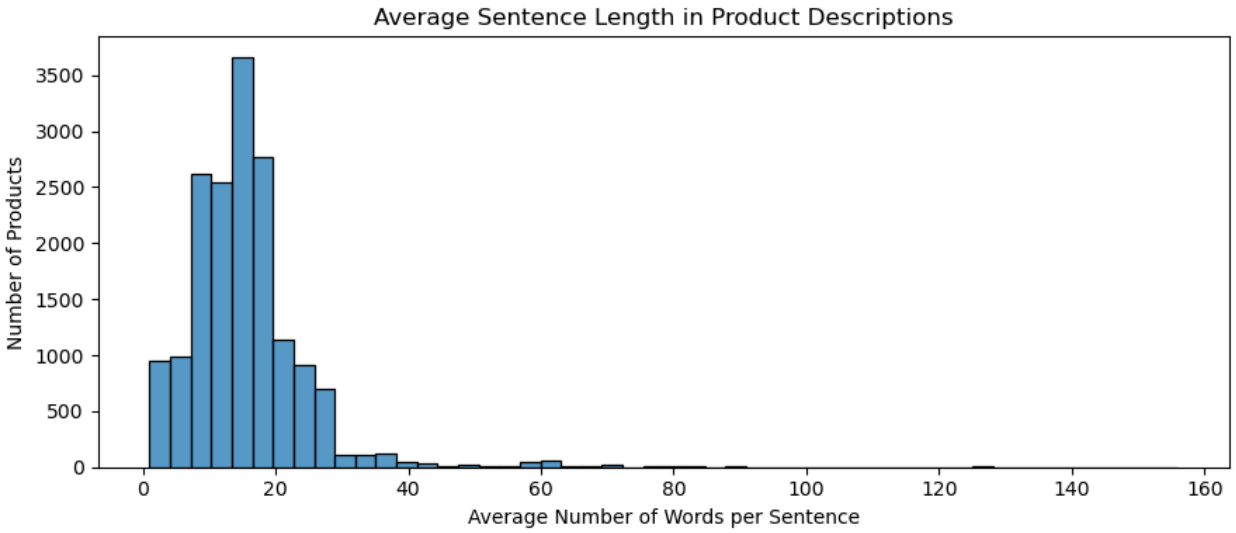
1. Nonempty descriptions have a sentence length of 10 to approximately 20 (18.67) words.
2. The mean and median are 15.42 and 14.43, respectively, and are relatively close.

Statistic	Value	Statistic	Value	Statistic	Value	Statistic	Value
count	16927.00	std	8.94	25%	10.00	75%	18.67
mean	15.42	min	1.00	50%	14.43	max	156.00

Summary statistics

As we can see from our distribution:

This distribution is right-skewed, which indicates that descriptions with longer sentences are rare.



2.4- Price Distribution:

First, we can notice from our graphs that both distributions are right-skewed. So, a small number of products have really high prices, while most of them have a low to middle price range.

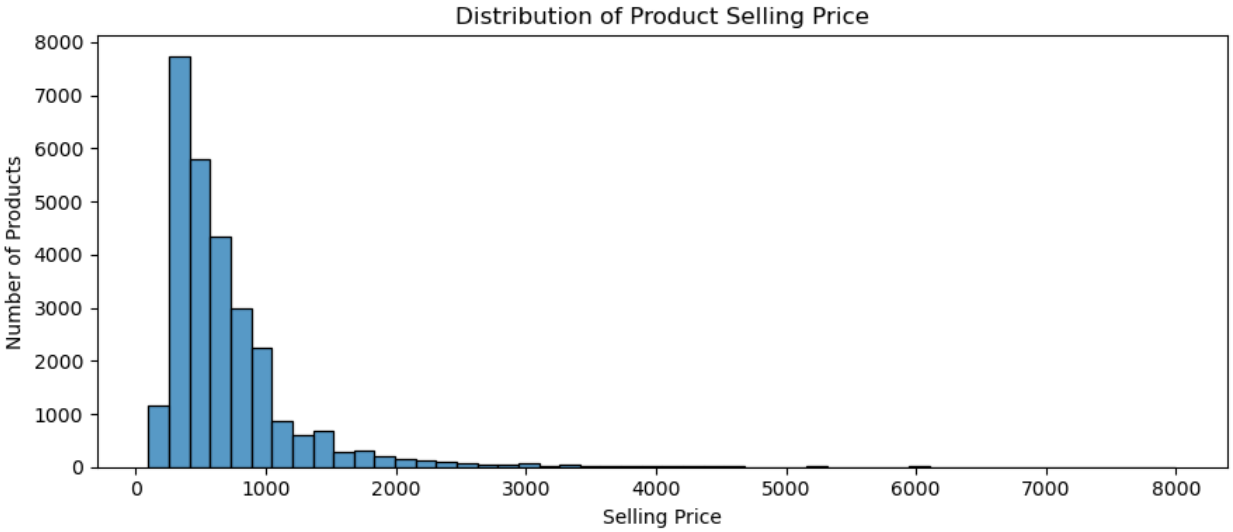
Second, we can also notice that selling prices are lower than actual prices, with a median of 545 for the first and 1199 for the second. This suggests that the dataset contains many discounted products.

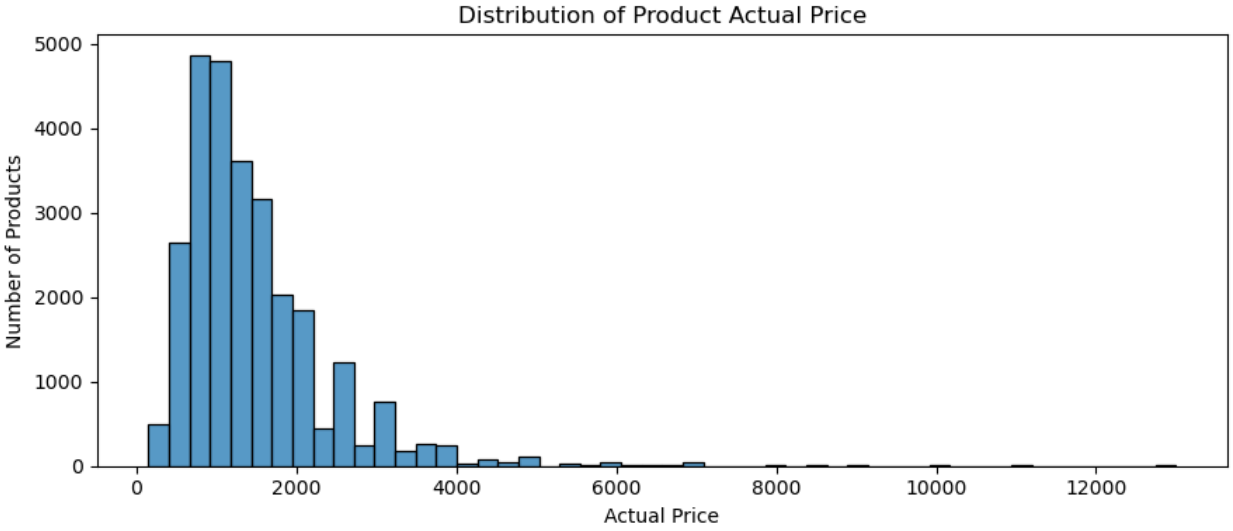
Third, the medians are lower than the means: A mean of 705.64 and a median of 545 for the selling price, and (1455.53;1199) for the actual price, in the same order as before.

This usually happens because some products are very expensive and they pull the averages upward.

	selling_price_num	**actual_price_num**
count	28078.0	27303.0
mean	705.64	1455.53
std	549.68	939.98
min	99.0	150.0
25%	390.0	849.0
50%	545.0	1199.0
75%	820.0	1799.0
max	7999.0	12999.0

Selling vs Discount Prices


Distribution of Product Selling Price
Number of Products
Selling Price
0
1000
2000
3000
4000
5000
6000
7000
8000


Distribution of Product Actual Price
Number of Products
Actual Price
0
2000
4000
6000
8000
10000
12000

2.5-Ratings and Discounts:

In this section, we analyzed product ratings and discount percentages to better understand these numerical attributes and how they could be useful in our search engine.

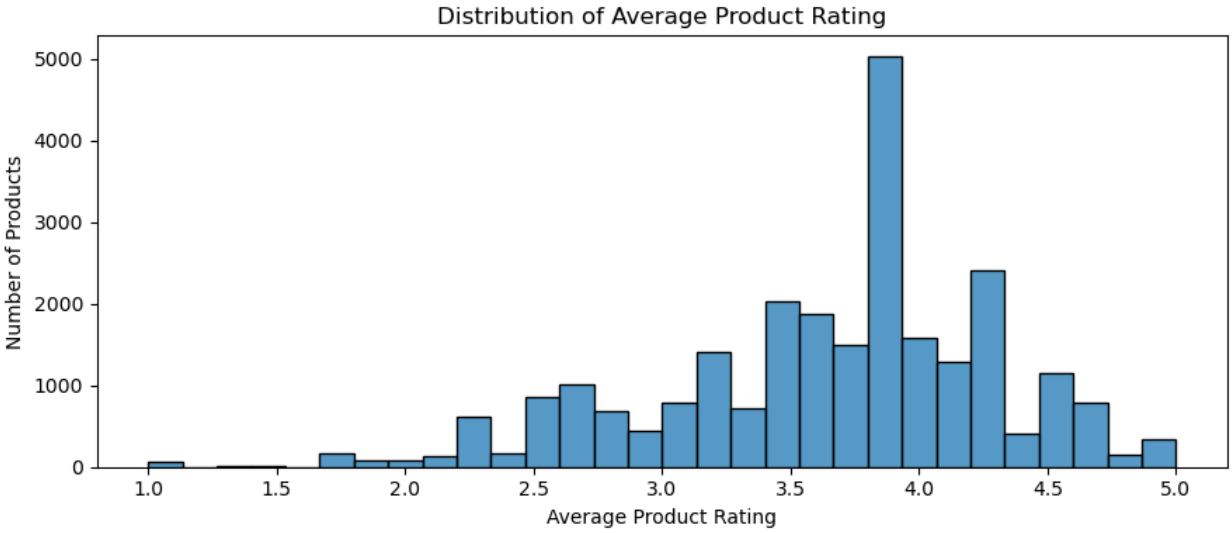
For product ratings, as we can see from our graphs, the distribution is concentrated between 3.5 and 4.5. (so towards higher values). The median rating is 3.8, which indicates that at least half of the products have a rating of 3.8 and above.

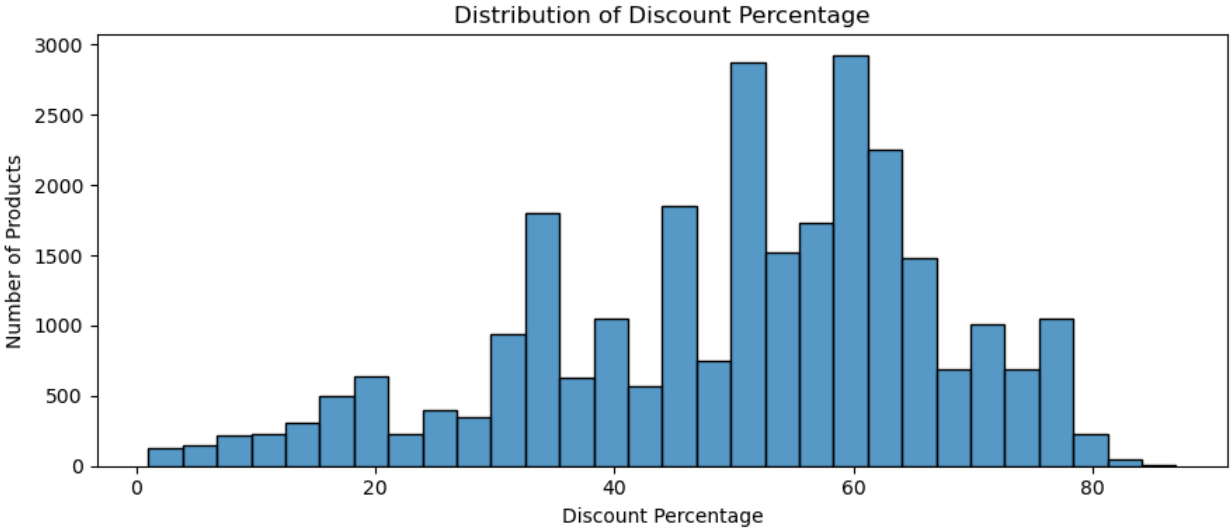
For discounts, as we can observe as well, the distribution is more widespread, with discounts ranging from 1% all the way up to 87%, but we can notice a concentration from 40 % to 65%, which suggests that many products have this discount percentage.

Additionally, from the table, we can find an average of 50.26% and a median of 53% for these discounted products.
Also, the reader can notice that we excluded missing values from statistical calculations. (2261 and 855 products have, respectively, missing ratings and discounts.

	average_rating_num	**discount_pct**
count	25819.00	27225.0
mean	3.63	50.26
std	0.66	16.89
min	1.00	1.0
25%	3.20	40.0
50%	3.80	53.0
75%	4.10	63.0
max	5.00	87.0

Ratings w.r.t. Discounts


Distribution of Average Product Rating
Number of Products
Average Product Rating
0
1000
2000
3000
4000
5000
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


Distribution of Discount Percentage
Number of Products
Discount Percentage
0
500
1000
1500
2000
2500
3000
0
20
40
60
80

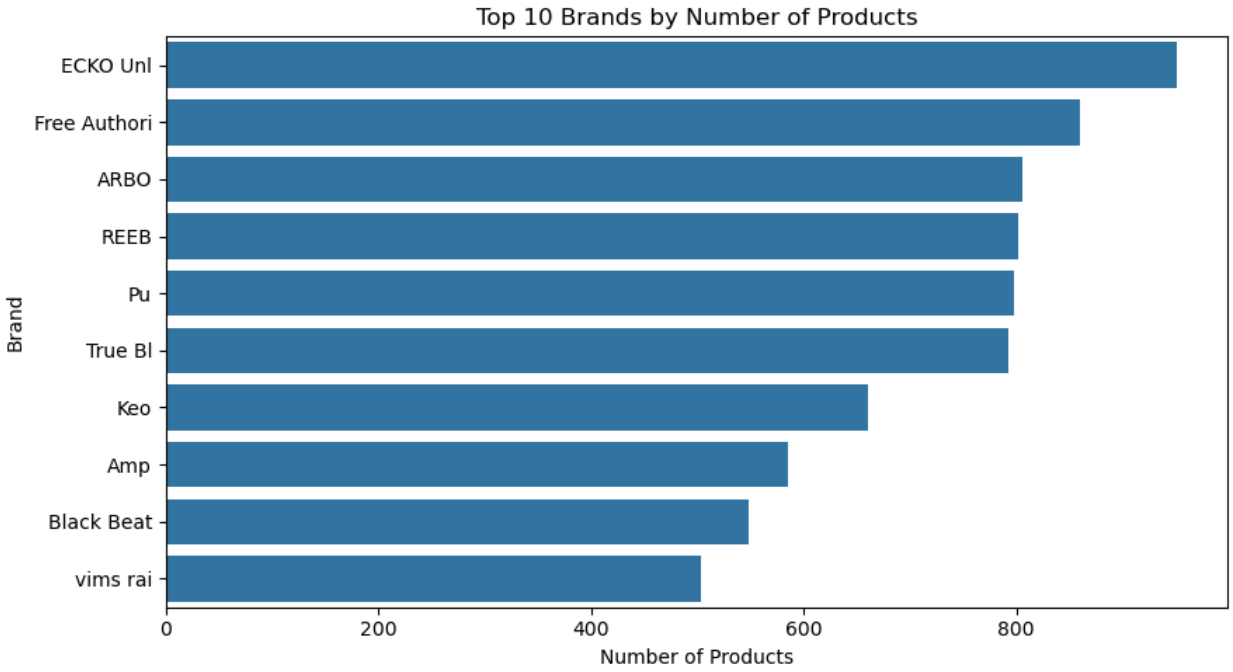
2.6- Top Brands and Sellers:

From observing our graphs and tables, we can draw three main conclusions:

- First, we can see that some brands and sellers are represented more than others, e.g., among brands, ECKO Unl has the largest number of products, while RetaiolNet has the most among sellers.
- Second, we can observe that 2,009 products have no recorded brand, while 1,643 have no seller information.
- Third, we should keep in mind that brand and seller information can be useful when a customer searches for a product associated with them (a product from this brand, or this seller). However, a brand or seller having more products doesn't mean they become relevant to all queries.

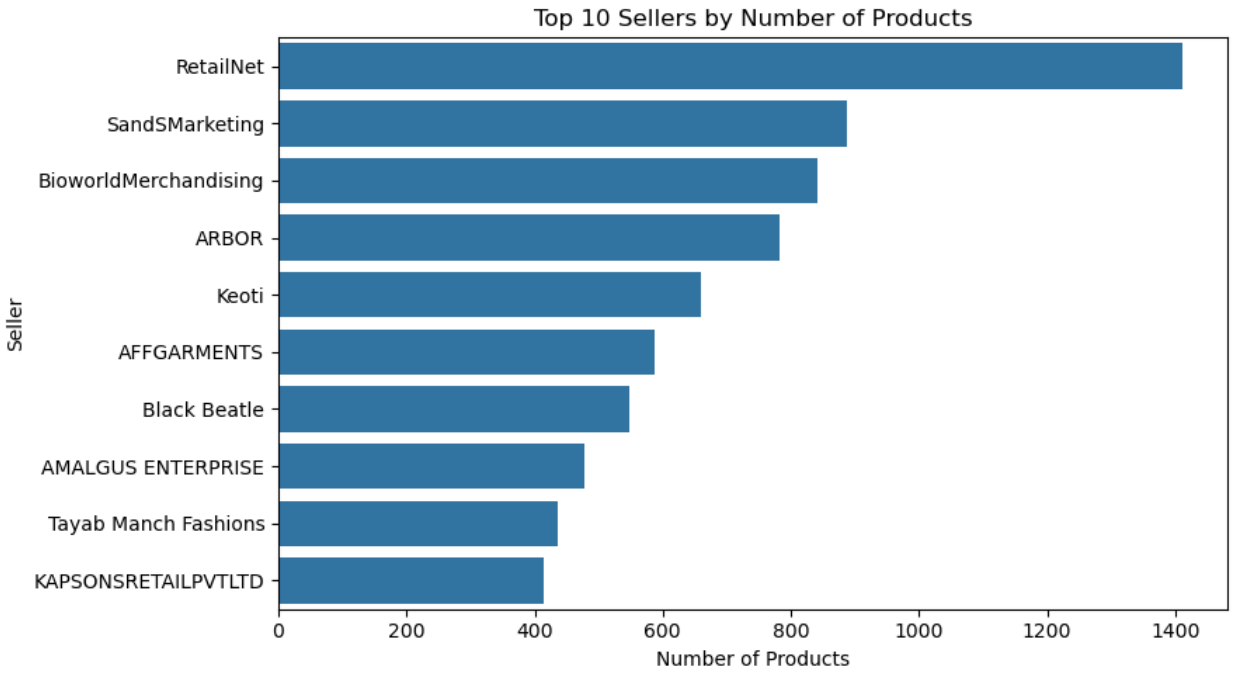
Brand	Frequency	Brand	Frequency
ECKO Unl	951	True Bl	793
Free Authori	860	Keo	660
ARBO	806	Amp	585
REEB	802	Black Beat	548
Pu	798	vims rai	503

Top 10 brands



Seller	Frequency	Seller	Frequency
RetailNet	1,411	AFFGARMENTS	587
SandSMarketing	887	Black Beatle	548
BioworldMerchandising	842	AMALGUS ENTERPRISE	477
ARBOR	783	Tayab Manch Fashions	436
Keoti	660	KAPSONSRETAILPVTLTD	415

Top 10 sellers



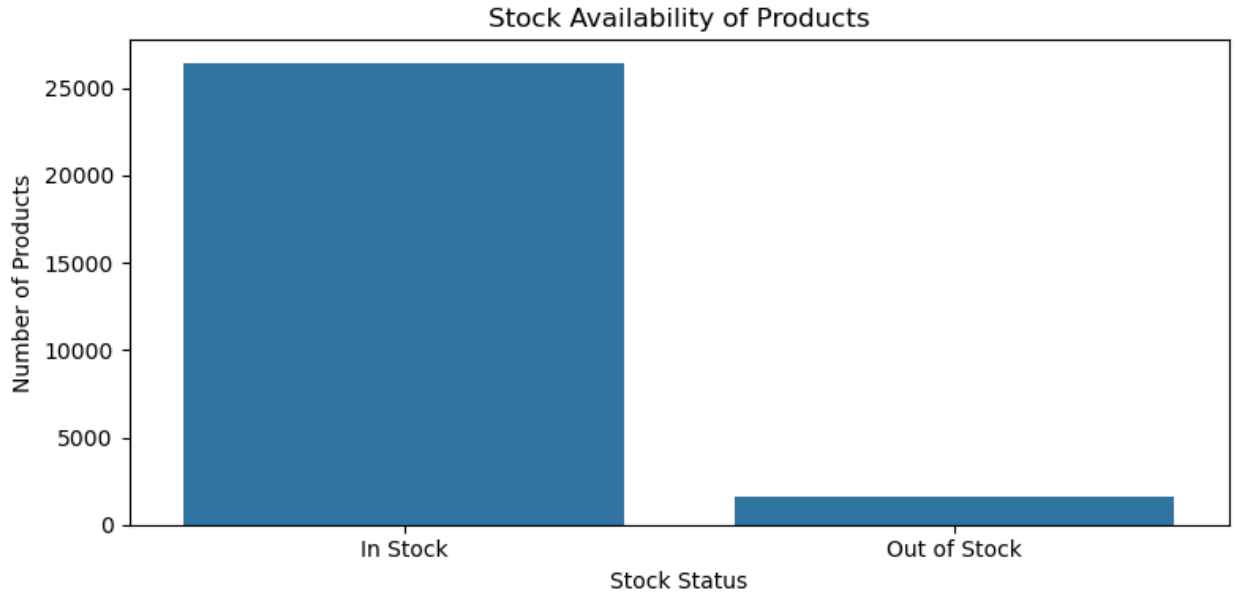
2.7- Stock Availability:

From an IR perspective, stock availability is important as it will affect the usefulness of search results. Effectively, a product might be interesting for a customer's search query, but if the product itself is unavailable, then it becomes automatically non-useful.

For this reason, we analyzed this stock availability to determine the proportion of products marked as available or unavailable in our dataset.

Therefore, we can notice from our visualizations that **6,436 products (94.15%)** are marked as in stock, while **1,644 products (5.85%)** are marked as out of stock. This indicates that, at the time the information was being collected, most of the products in our dataset were available.

Status	**Number of Products**	**Percentage (%)**
In Stock	26436	94.15
Out of Stock	1644	5.85



2.8- Category and Subcategory Distribution

In this section, we analyzed the distribution of products across categories and subcategories to better understand the composition of our dataset.

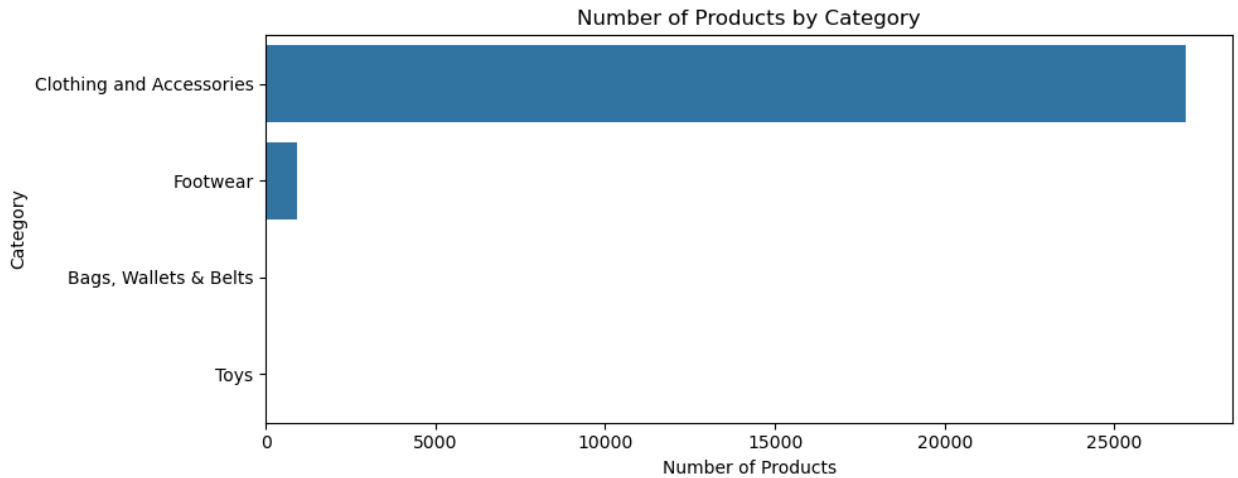
As we can observe, **Clothing and Accessories** is by far the most represented category with 27118 products, then comes footwear with 920 products while Bags, Wallets & Belts and Toys represent only a very small proportion of the dataset.

From the subcategories, we can see that Top wear alone represents 54.52% of all products, and the three largest subcategories (Top wear, Bottom wear, and Winter Wear) account for approximately 76.46% of the dataset.

This analysis is important because it gives further proof that indexing different fields is necessary, because when a category contains thousands of products, general terms from the categories and subcategories with the greatest percentages become less discriminative for our search engine. Therefore, we need more specific terms that describe the type, material, size, fit, etc., of any product.

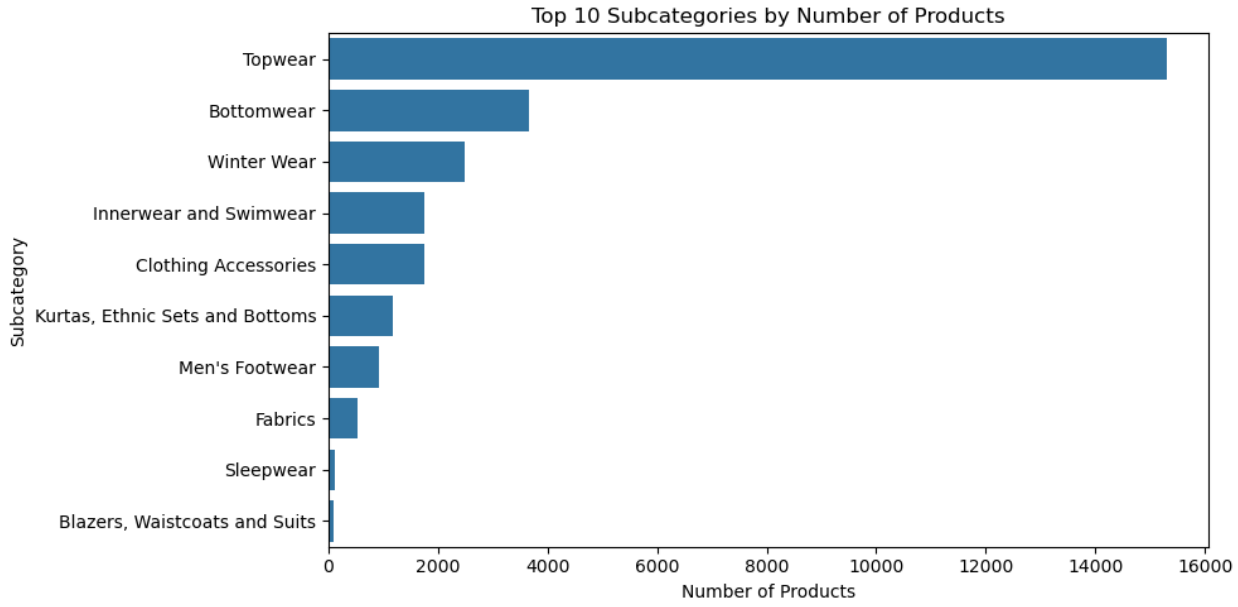
Category	Count	Category	Count
Clothing and Accessories	27,118	Bags, Wallets & Belts	41
Footwear	920	Toys	1

Products by category



Subcategory	Count	Subcategory	Count
Topwear	15,310	Kurtas, Ethnic Sets and Bottoms	1,180
Bottomwear	3,666	Men's Footwear	920
Winter Wear	2,493	Fabrics	534
Innerwear and Swimwear	1,762	Sleepwear	122
Clothing Accessories	1,742	Blazers, Waistcoats and Suits	103

Top 10 subcategories



2.9-Word Clouds:

From the word clouds we generated, we can notice that the most frequent words in the titles and the product descriptions include, respectively:

"shirt" (describes product type)

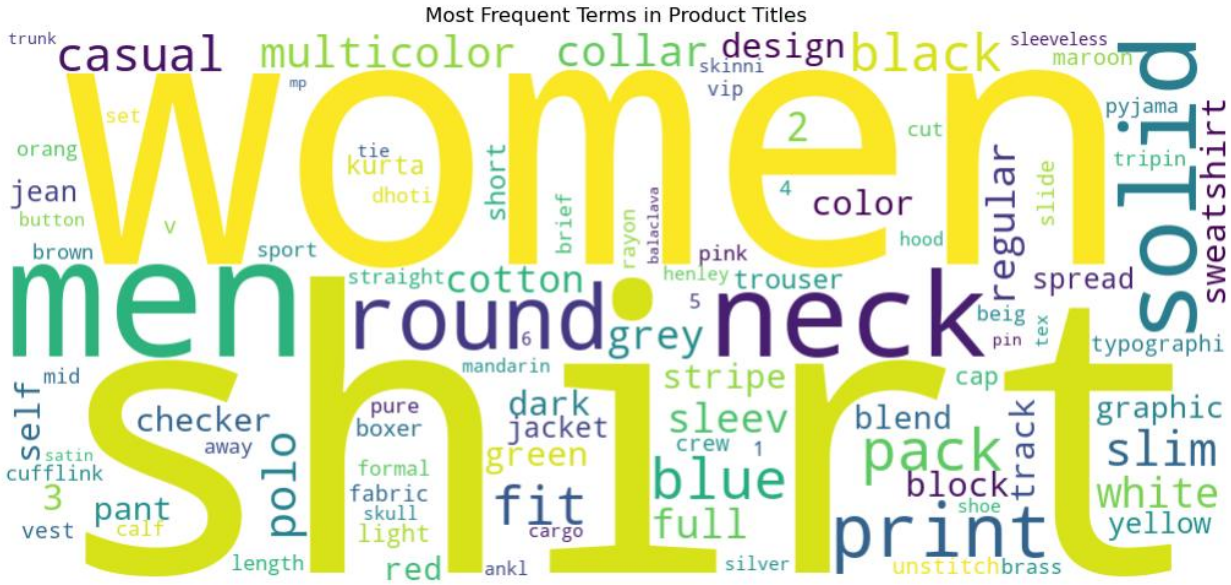
"women, men" (describe intended audience)

"neck, solid and round" (describe clothing styles)

"shirt, cotton", (information about materials)

"comfort, wear, fabric, fit" (describe product characteristics)

Importantly, titles identify what the product is, and descriptions provide additional details about its characteristics.



Most Frequent Terms in Product Descriptions



2.10 Product Ranking:

We ranked the products based on their average ratings, selling prices, and discount percentages, to check the products with the highest ranking in each category.

- Average Rating:

  The highest-rated products all had a rating of 5 out of 5, including men's briefs, women's vests, T-shirts, and casual shirts. But since many products have this rating, we can conclude that these five products are the best among all.

- Selling prices:

  The most expensive product is Solid Woman Track Suit (priced for 7,999).

- Discount Percentages:

  The highest discount is **87%**, corresponding to a *Printed Women Round Neck White T-Shirt*. Also, 4 out of the 5 most discounted products are T-shirts.

All of this shows that numerical statistics can be used to rank products according to different criteria; however, this does not mean that these products will necessarily be relevant to a search query. Then, we decided to keep these statistics outside our ranking data or textual inverted index, so that this data can support possible filtering and sorting features in the future.

	Title	**average_rating_num**
22361	Men Brief (Pack of 5)	5.0
8403	VIP Women Vest (Pack of 8)	5.0
14764	Graphic Print Women Round Neck White T-Shirt	5.0
13240	Men Brief (Pack of 5)	5.0
10472	Men Slim Fit Printed Spread Collar Casual Shirt	5.0

Top 5 highest-rated products

	Title	selling_price_num
2067	Solid Women Track Suit	7999
11010	Checkered Single Breasted Party Women Full Sle...	7998
26089	Full Sleeve Printed Men Sweatshirt	7799
25815	Full Sleeve Solid Women Casual Jacket	7799
6895	Skinny Men Blue Jeans	7794

Top 5 most expensive products

	Title	discount_pct
906	Printed Women Round Neck White T-Shirt	87
903	Printed Women Round Neck White T-Shirt	86
902	Printed Women Round Neck Grey T-Shirt	86
18249	Printed Men Round Neck Multicolor T-Shirt (Pa...	85
9811	Solid Balclava Cap	84

Top 5 most discounted products